Homework 03

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Part 1: Setting up tasks

Reading in packages

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)
```

Creating data object

```
# creating new object for temp-kelp data
kelp <-
  # reading csv
  read_csv(
    # directing function to data file
    here("data", "temp-kelp.csv"))

# creating new object for personal data
job_search <-
  # reading csv
  read_csv(
    # directing function to data file
    here("data", "personal-data.csv"))
```

Part 2: Problems

Problem 1: Giant kelp fronds

Climate change threatens all ecosystems, including giant kelp (Macrocystis pyrifera) forests.

You are interested in the potential relationship between ocean temperature (measured in ° C) and giant kelp frond elongation rate (also known as growth rate, measured in cm day-1).

You conduct a field study in which you track giant kelp individuals growing in different temperatures (on average) and measure their elongation rates. Admittedly, this is not a perfect study, but you do the best with what you have!

1a. An appropriate test

In 1-3 sentences, name the appropriate test(s) to determine the strength of the relationship between temperature and giant kelp frond elongation rate (hint: there are two). Describe the differences between the two tests.

Be specific in your response to demonstrate your understanding of the variables in this question.

Because I solely want to understand the strength of the relationship between temperature and giant kelp frond elongation rate, I want to use a correlation test.

A Pearson's r is used when there is a linear relationship between two variables. A Spearman's rho is used when the relationship between two variables is monotonic.

1b. Create a visualization

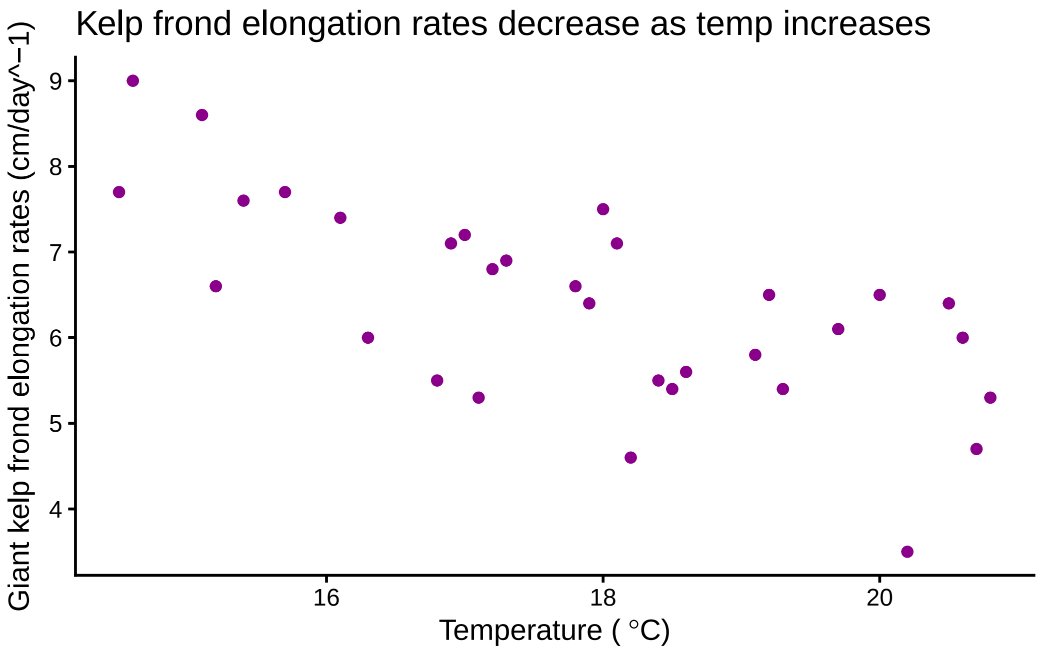
Create a visualization that would be appropriate for showing the relationship between temperature and giant kelp frond elongation rate.

In addition to using the correct geometries, be sure to:

- ***relabel the x- and y-axes and include units***
- ***use different colors from the ggplot() defaults***
- ***use a different theme from the ggplot() default***

```r
# creating a new object
temp_kelp_visual <-
  # base layer: ggplot
  ggplot(
    # df: kelp
    data = kelp,
    # x-axis: temp_c
    # y-axis: kelp_elong
       mapping = aes(x = temp_c,
                     y = kelp_elong)) +
  # first layer: scatterplot
  geom_point(
    # changing color of points
    color = "magenta4") +
  # changing theme
  theme_classic() +
  # changing title and axes' titles
  labs(title = "Kelp frond elongation rates decrease as temp increases",
       x = expression("Temperature ("*~degree*C*")"),
       y = expression("Giant kelp frond elongation rates (cm/day^-1)"))

# displaying object
temp_kelp_visual
```

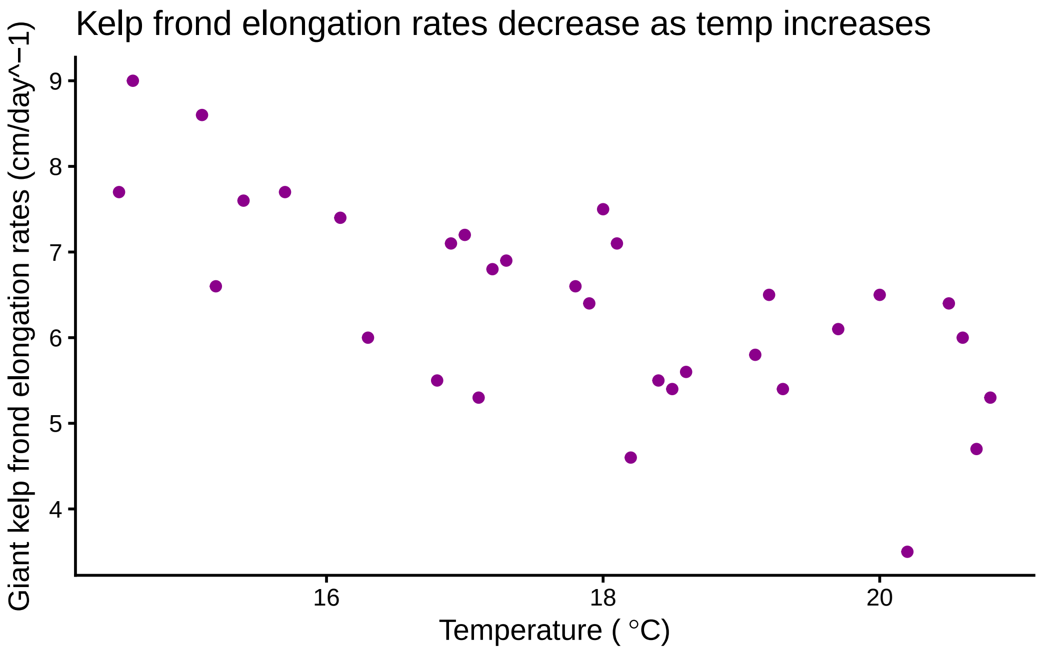


1c. Check your assumptions and run your test

Assumptions

To run a Pearson's r, you want to ensure that there is a linear relationship between the two variables. From the scatter plot I created in part 1b, I can conclude that the relationship mostly appears to be linear.

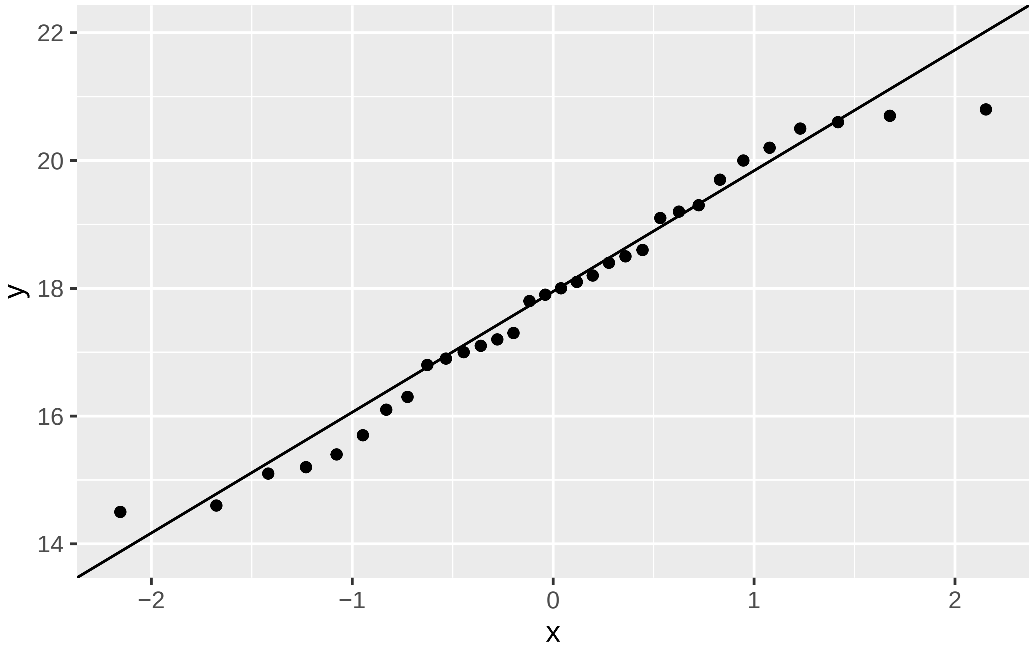
```
# displaying object from 1b
temp_kelp_visual
```



I also want to make sure that the data is normally distributed. For this, I can create a QQ plot.

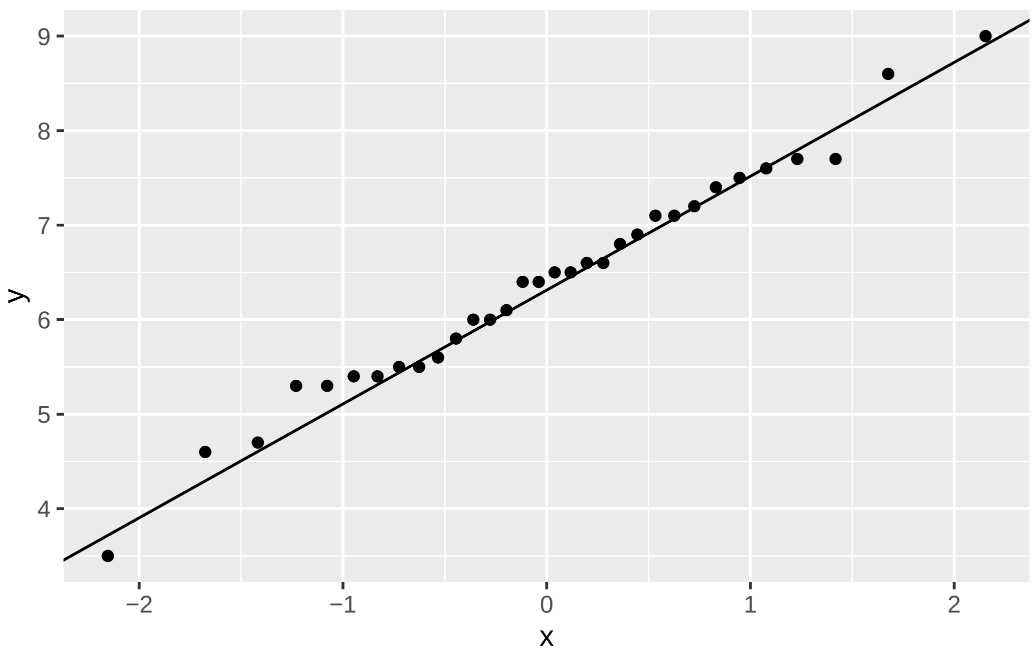
```
# creating new object
qqplot_temp <-
  # baselayer: ggplot
  ggplot(
    # df: kelp
    data = kelp,
    # variable: temp
       aes(sample = temp_c)) +
  # first layer: distribution of temp observations
       stat_qq(distribution=qnorm) +
  # second layer: expected distribution of temp
stat_qq_line( col = "black")

# displaying object
qqplot_temp
```




```r
# creating new object
qqplot_kelp <-
  # first layer: ggplot
  ggplot(
    # df: kelp
    data = kelp,
         # variable: kelp_elong
      aes(sample = kelp_elong)) +
  # first layer: distribution of kelp elongation rate observations
  stat_qq(distribution=qnorm) +
  # second layer: expected distribution of kelp elongation rate
  stat_qq_line( col = "black")

# displaying object
qqplot_kelp
```



It seems that the variables are mostly normally distributed.

Additionally, the variables of kelp frond elongation rates and temperature need to be continuous, which they are since they are not integers, categorical, or ordinal data.

Finally, the observations also need to be independent, which is true because the observations of kelp growth rates do not all come from the same individual.

All of our assumptions have been checked so we can now run the Pearson's r.

Pearson's r test

```
#correlation test
cor.test(
  # df: kelp
  # first variable: temp
  kelp$temp_c,
  # df: kelp
  # second variable: kelp elongation rates
         kelp$kelp_elong,
  # type of correlation test: Pearson's r
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

1d. Results communication

In 1-3 sentences each, write about:

- ***Which test you used, and why (i.e. "To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a…")***
- ***Your interpretation of your test (along with the appropriate summary of the test in parentheses)***

To analyze the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Pearson's r correlation test.

My results suggest that I can reject the null hypothesis, and that there is a moderate negative relationship between temperature and giant kelp frond elongation rate (Pearson's r = -0.69, t(30) = -5.2, $p < 0.001$, $\alpha = 0.05$).

1e. Test implications

You're working on a team of people who are also interested in the potential relationships between temperature and giant kelp growth rate, and more broadly giant kelp health. In 2-3 sentences, write what you would communicate to them about the results of this test and what it means for giant kelp growth.

Be cognizant of your audience as you are writing: what would they need to know to take action?

I found a moderate negative (Pearson's r = -0.69) relationship between temperature and giant kelp frond elongation rate. Elongation rates are highest when temperature is below 16 °C and begins to decrease from there. While other factors may influence the overall health of giant kelp, the moderate correlation between temperature and frond elongation rates suggests that understanding changes in temperature and overall climate will be a significant factor in monitoring giant kelp growth.

1f. Double check your own work

In part a, you outlined two potential tests to answer this question about the strength of the relationship between temperature and giant kelp front elongation rate. In part c, you chose a test, checked your assumptions, and ran one.

Try running the other test you listed in part a. Include the annotated code and output.

```
#correlation test
cor.test(
  # df: kelp
  # first variable: temp
  kelp$temp_c,
  # df: kelp
  # second variable: kelp elongation rates
         kelp$kelp_elong,
  # type of correlation test: Spearman's rho
         method = "spearman")
```

```
    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

In 1-3 sentences, describe whether or not the two tests would have led you to make the same decision (about the null hypothesis) and interpret the results the same way (about the strength of the relationship between temperature and giant kelp front elongation rate).

In your description, be specific about the tests, their components, and their relation to the variables.

Both the Pearson's r and the Spearman's rho correlation tests would have led me to reject the null hypothesis that there is no correlation between temperature and kelp frond elongation rates, because they both had a significant p-value that was less than the significance level of $\alpha = 0.05$.

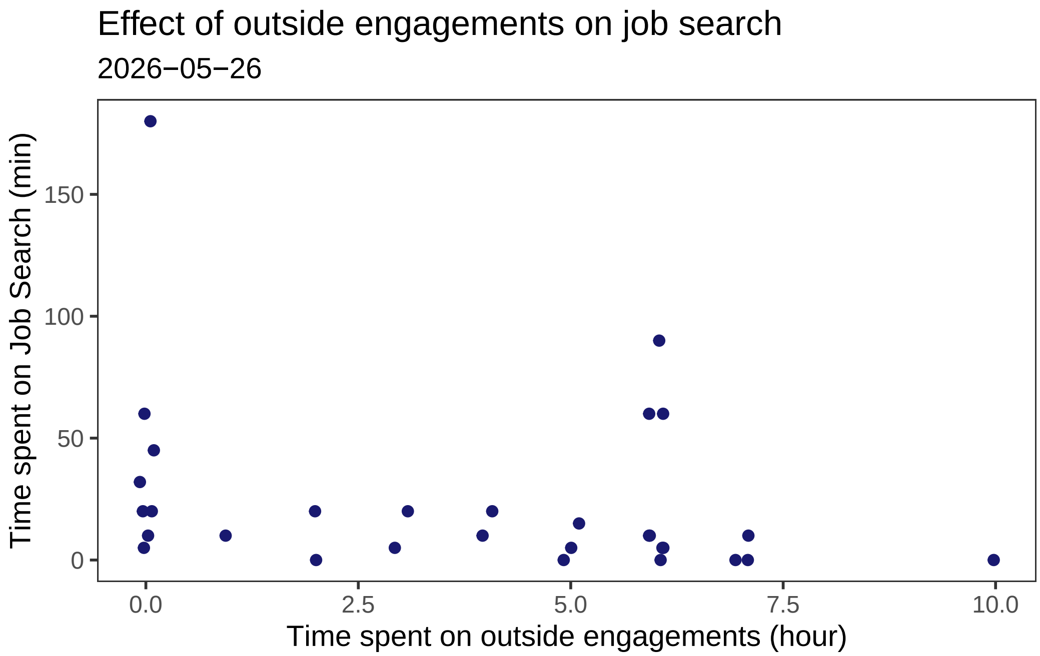
Additionally, the resulting r and rho was the same, -0.69, so both tests would lead for me to determine that there is a moderate negative correlation between temperature and kelp frond elongation rates. Both tests result in the same r and rho value because the variables they meet the assumptions of both the Pearson's r and the Spearman's rho.

##Problem 2: Personal data

2a. Updating visualizations

Effect of outside engagements on job search

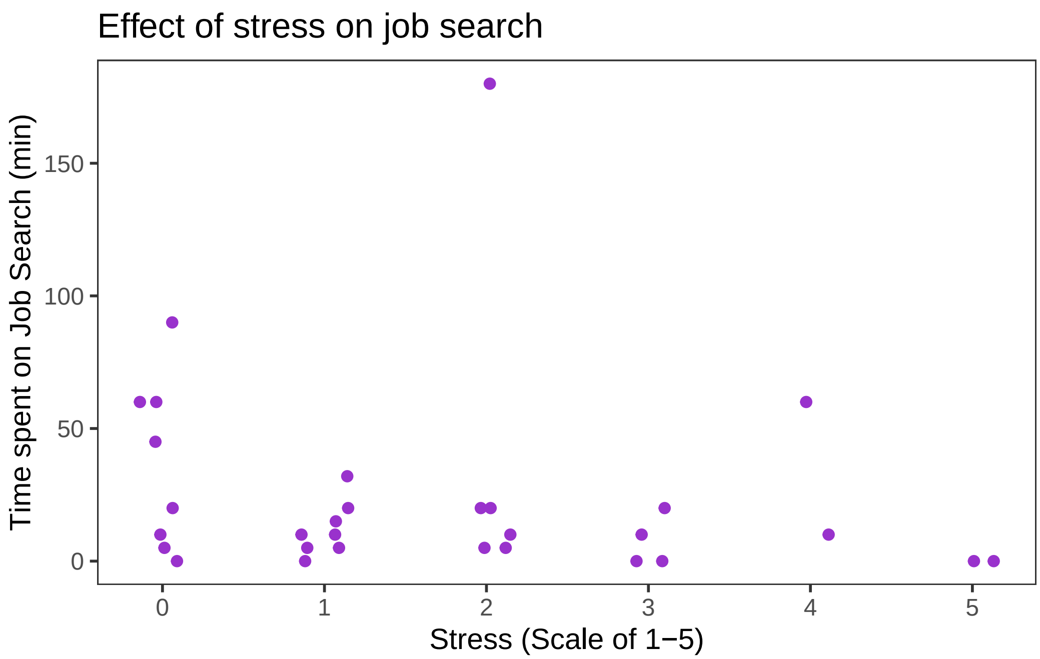
```
# creating new object
outside_eng_as_predictor <-
  # first layer: ggplot
  ggplot(
    # df: job_search
    data = job_search,
       mapping = aes(
         # x-axis: outside engagements
         # y-axis: time spent on job search
         x = outside_eng,
         y = job_search)) +
  # first layer: points
geom_jitter(color = "midnightblue",
            height = 0,
            width = 0.1) +
  # changing the title and axes' titles
  labs(title = "Effect of outside engagements on job search",
       subtitle = "2026-05-26",
       x = "Time spent on outside engagements (hour)",
       y = "Time spent on Job Search (min)") +
  # changing the theme
  theme_bw() +
  # getting rid of visual clutter
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank())
# displaying the plot
outside_eng_as_predictor
```



Effect of stress on outside engagements

```
# creating a new object
stress_as_predictor <- ggplot(data = job_search,
# x-axis: level of stress
# y-axis: time spent on job search
       mapping = aes(x = stress,
                     y = job_search)) +
  # first layer: points
geom_jitter(color = "darkorchid",
            height = 0,
            width = 0.15) +
  # changing title and label of axes
  labs(title = "Effect of stress on job search",
       x = "Stress (Scale of 1-5)",
       y = "Time spent on Job Search (min)") +
  # changing the theme
  theme_bw() +
  # getting rid of visual clutter
    theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank())
# displaying the plot
```

```
stress_as_predictor
```



2b. Caption writing

Figure 1: Time spent on job search differs with varying time spent on outside engagements. Small dark blue points represent an observation of recorded time spent on searching and applying for jobs in a single day. Points are distributed across x-axis, which represents the hours spent on outside engagements, including school (lectures) and work (shifts).

Figure 2: Time spent on job search differs with varying levels of stress. Small purple points represent an observation of recorded time spent on searching and applying for jobs. Points are distributed across x-axis, which represents the the level of stress recorded on that day.

Note: I changed the stress column so that it would be categorical, with a value of 0 meaning I have no assignments or chores (with chores including necessary appointments, extracurricular, or meetings that are not work or school related). - 0: No assignments or chores (chores including chores, necessary appointments/meetings, or extracurricular activities)

- *1: One assignment or chore*
- *2: Two of either assignments or chores or a mix of both*
- *3: Three of either assignments or chores or a mix of both*

- *4: Midterm on that date*
- *5: Midterm plus one or more of an assignment or chore*